

Diploma in

PHOTOSHOP



SUMMARY NOTES

Sharpening & Output
Lessons 29 & 30

THE LESSON WILL DISCUSS SCREEN CALIBRATION, PRINTER PROFILING, COLOUR SPACES, AND HOW TO SHARPEN CORRECTLY FOR BOTH SCREEN AND PRINT. THESE SUBJECTS ARE ESSENTIAL TO THE CONSISTENT AND PREDICTABLE OUTPUT OF YOUR IMAGES.

3.01	Colour Space
3.02	Colour Calibration
3.03	Colour Correction
3.04	Sharpening Filters

Sharpening & Output

COLOUR SPACE

“Vectors are now a big part of Photoshop creations and can be used to make various styled illustrations.”

A device color space simply describes the range of colors, or gamut, that a camera can see, a printer can print, or a monitor can display. Editing color spaces, on the other hand, such as Adobe RGB or sRGB, are device-independent. They also determine a color range you can work in. Their design allows you to edit images in a controlled, consistent manner.

In layman's terms, a colour space is just a specific range of colours that we can use to illustrate our photo. A narrow colour space will have a limited amount of colours it can draw from to make up a photo, while a wide colour space will have a much higher number of colours it can draw from the make up our photo.

Our chosen colour space really depends on the purpose of the file. So, an image for print would probably need a different colour space that what we would need for screen

Although there are many different colour spaces, the 3 to mainly focus on and understand are:

- sRGB – Standard RGB
- Adobe RGB 1998
- ProPhoto
- CMYK

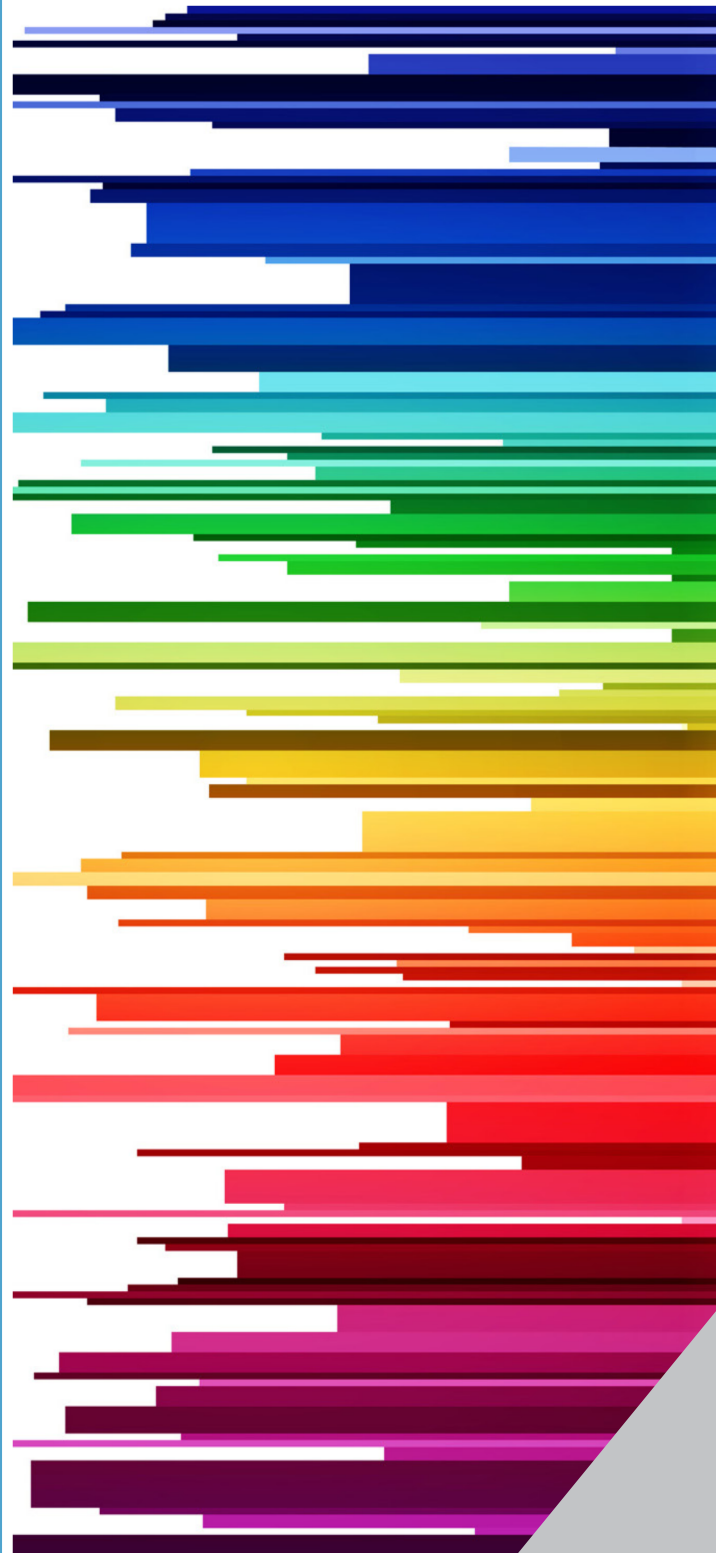
sRGB

We come in contact with RGB every single day, on our computer screens, our mobile devices and all over the Internet. It's basically the jpeg of colour spaces. In fact, most jpegs out there are sRGB and this is because sRGB is a standard RGB colour space created cooperatively by HP and Microsoft in 1996 for use on monitors, printers and the Internet. sRGB is a great way to guarantee that your photos look great on the web.

Note: Traditional computer monitors can only display about 97% of the sRGB colour space.

Adobe RGB 1998

Adobe RGB represents a wider range of possible colours using the same amount of information as sRGB by making the colours more spaced out, deeper, richer. Since sRGB has a narrower range of colours than Adobe RGB, it cannot display certain highly saturated colours that could still be useful in certain applications, such as professional-grade printing. Thus, photographers and graphic artists that need this extra colour range for specific purposes would choose Adobe RGB over sRGB. We can always convert back to sRGB if we need to down the line. Unfortunately, Adobe RGB isn't great for the web and if you use it, your colours will be displayed inaccurately.



Pro-Photo

Pro-Photo RGB is just about the widest colour space. However, because it is so wide, it can give you all kinds of grief if you don't know exactly what you are doing. It contains many, many, MANY more colours than you can see on any monitor, it even encompasses what can only be termed "theoretical colours" because they're outside the visible colour range in the physical world. A screen can't handle pro-photo, so you can only see the benefit when you are printing. In fact, on screen it doesn't look too good at all! It's used a lot in the US, whereas Europe tends to sway more to Adobe RGB 1998.

CMYK

CMYK is a very narrow colour gamut or workspace. CMYK is Cyan, Magenta, Yellow and Key. CMYK is reserved for print. This is because there is a fundamental difference between creating colours with light vs. simulating colours with ink. While printers use tiny dots containing only one of four colours (Cyan, Magenta, Yellow or Black) to create the illusion of other colours, screens are capable of producing millions of colours.

Bit Depth

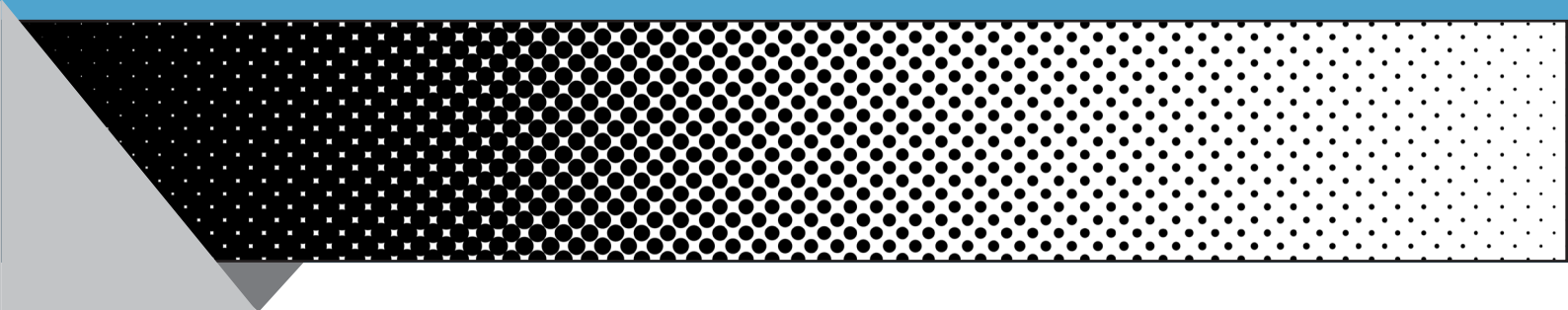
Bit depth does not influence the range of colours represented, it only influences the granularity between colours. So, sRGB/16 displays the same range of colours as sRGB/8, it can just display finer nuances and smoother gradients.

8 bit:

256 shades per channel 16.8 million colours

16 bit:

65,536 shades per channel 281 trillion colours



COLOUR CALIBRATION

The purpose of calibration is to make sure that your eye sees the photo the same way on your screen as others will see it. For printing, you can be sure that by calibrating your screen your photo will look very close to how it looked on your computer screen. However, when you edit on a calibrated screen and then post your photo online, it does not mean that everyone else will see the photo properly. Their screens are most likely uncalibrated, but it is still important to calibrate. Why? Because computer monitor manufacturers strive to make their products reproduce colours properly, so by colour calibrating, you'll be at neutral even if some screens are off one way and others are off another way.

To calibrate our screen, we use a device called a Colorimeter and it's a device that attaches directly to the front of your display and then reads a variety of colour values presented by special software:

- Colour munki
- Spyder
- X-rite

This process will result in an ICC profile for your display and that profile will be used to compensate for the behaviour of the display to make sure that the colours presented are accurate. Today's displays are quite stable and so you don't need to calibrate on too regular a basis. Every six months is, really, probably fine for most photographers, but you do need to do it at least initially.

In colour management, an ICC profile is a set of data that characterizes a colour input or output device, or a colour space, according to standards promulgated by the International Colour Consortium (ICC). And that's especially important because today's displays are also very bright. In fact, most LCD displays are about twice as bright as they really need to be and going through a monitor calibration and profiling process will help compensate for those issues.

However, almost without exception, computer monitors are kept brighter than a calibrated screen. So, if I edit a photo on a calibrated device, it will likely show up a TINY BIT darker on uncalibrated monitors.

COLOUR CORRECTION

Colour Correction Steps1

1. Add Hue & Saturation Non destructive adjustment. (This will allow you to take note what colours are exposed by temperature or natural light)
2. Add Colour Balance
3. Edit Sliders to suit your image.

Colour Correction Steps2

1. Menu>Window>Info>Use Info Panel (This will allow you match RGB values of image)
2. Add Colour Balance
3. Edit Sliders to suit your image.

Colour Correction Steps3

1. Duplicate layer(This will allow you match RGB values of image)
2. Add Filter> Blur > Average
3. Add Levels or Curves > use gray point dropper



SHARPENING FILTERS

- Sharpen and Sharpen More - Focus a selection and improve its clarity. The Sharpen More filter applies a stronger sharpening effect than does the Sharpen filter.
- Sharpen Edges and UnsharpMask - Find the areas in the image where significant colour changes occur and sharpen them. The Sharpen Edges filter sharpens only edges while preserving the overall smoothness of the image. Use this filter to sharpen edges without specifying an amount. For professional colour correction, use the Unsharp Mask filter to adjust the contrast of edge detail and produce a lighter and darker line on each side of the edge. This process emphasizes the edge and creates the illusion of a sharper image.
- Smart Sharpen - Sharpens an image by letting you set the sharpening algorithm or control the amount of sharpening that occurs in shadows and highlights. This is the recommended way to sharpen if you don't have a particular sharpening filter in mind. See Sharpen using Smart Sharpen. In Photoshop CC, the enhanced Smart Sharpen filter empowers you to produce high-quality results through adaptive sharpening technology that minimizes noise and halo effects. The streamlined UI design for this filter offers optimized controls for targeted sharpening. Use the sliders for quick adjustments and advanced controls to fine-tune your results. Smart sharpening in Photoshop CC supports CMYK. Additionally, you can sharpen arbitrary channels. For example, you can choose to sharpen just the blue channel, green channel, or the alpha channel.

Here is a recommended approach to sharpening images with Photoshop CC:

- To begin, set Amount to a high value.
- Increase the Radius to a value that introduces a halo effect.
- Decrease the Radius until the halo effect disappears. You've arrived at the optimal value for Radius.
- Now, decrease the Amount value as necessary.
- Adjust the Reduce Noise slider, such that the noise in the image looks similar to how it was before you began sharpening the image. Too much noise reduction can result in a plastic look. Higher Amount values require greater noise reduction





THANK YOU